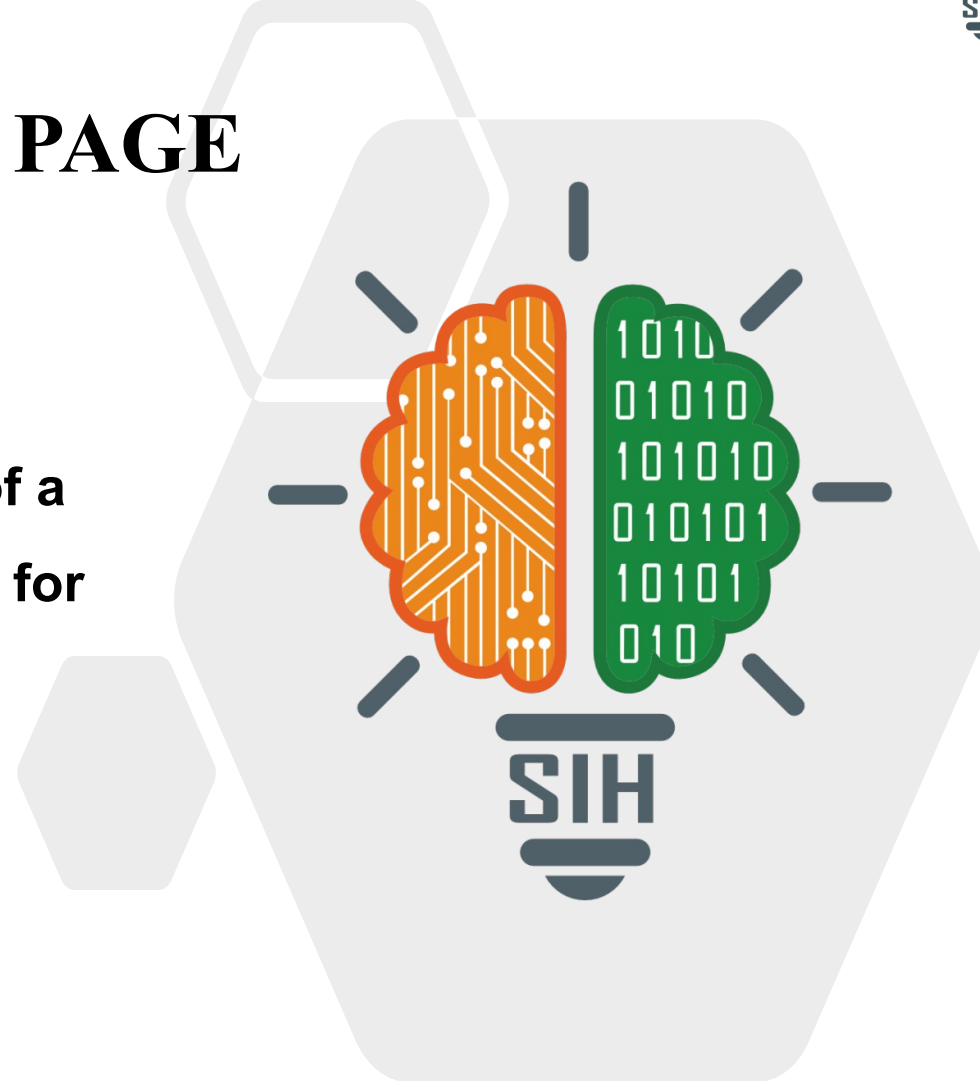
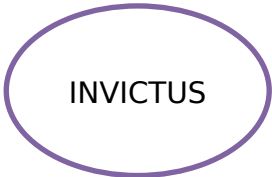




## TITLE PAGE

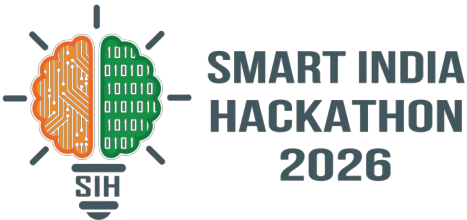
- Problem Statement ID – 26110
- Problem Statement Title- Development of a Low-Cost Light-weight Milk Chilling Can for Small-Scale Dairy Farmers
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Hardware
- Team ID- 159400
- Team Name- INVICTUS



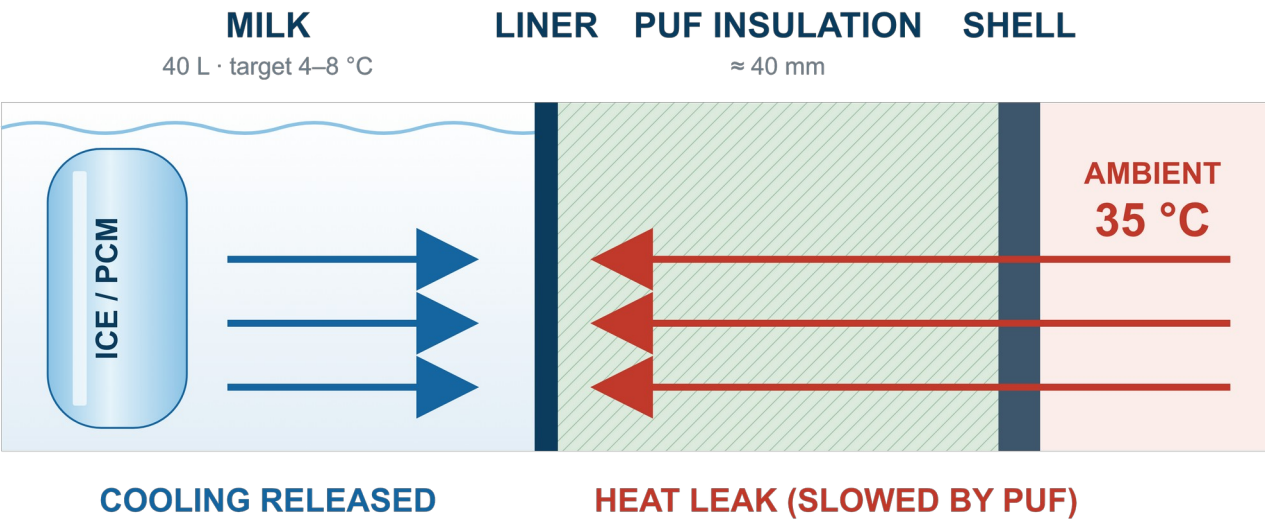


# MILKSHIELD

Passive Cooling Milk Chilling Can, 30–40 L



## ❖ Proposed Solution



|                            |                                   |                            |                                |
|----------------------------|-----------------------------------|----------------------------|--------------------------------|
| <b>30–40 L</b><br>capacity | <b>4–8 °C</b><br>milk temperature | <b>6–12 h</b><br>hold time | <b>0 W</b><br>continuous power |
|----------------------------|-----------------------------------|----------------------------|--------------------------------|

PS requirements – simulated: below 8 °C in 75 min, held 14 h at 35 °C

Optional monitoring:



### WHAT IT IS

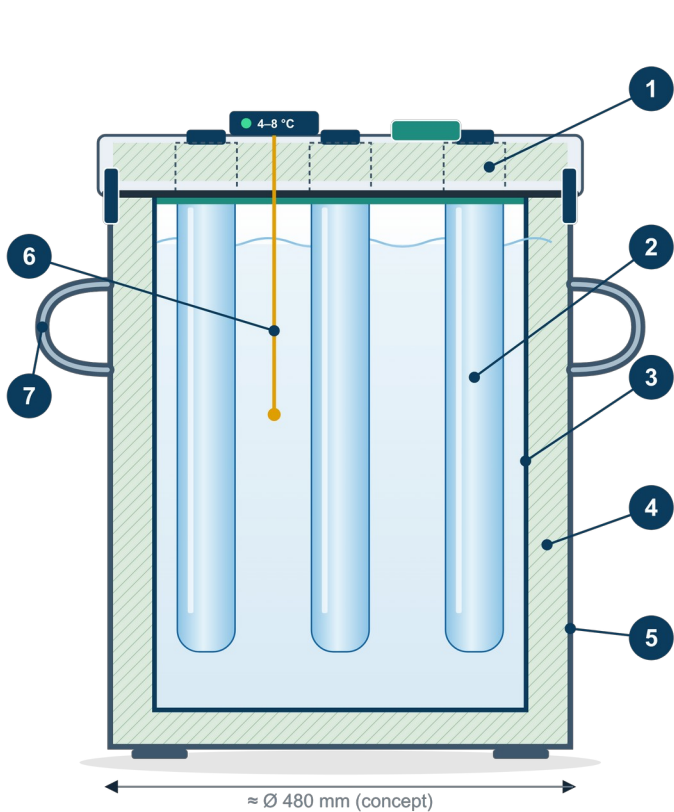
- Double-wall PUF-insulated can with a food-grade HDPE chamber
- Reusable ice/PCM cartridges sit in the milk – no compressor, no mains power

### HOW IT WORKS

*Passive cooling stores cooling energy in reusable PCM/ice cartridges and releases it gradually around the insulated milk chamber without continuous external power.*

### WHAT IS NEW

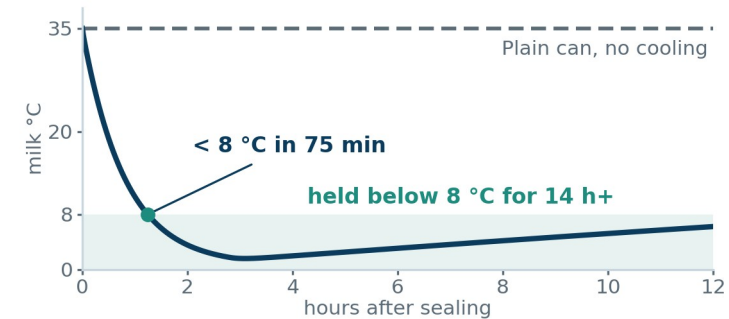
- **Modular cartridges:** capacity set per load
- **Ice inside the milk:** direct, fast heat transfer
- **Easy to clean:** smooth, wide-mouth liner



- Lid + gasket**  
Insulated PE lid, food-grade seal, cam latches
- Ice/PCM cartridge**  
Sealed 2 L food-grade HDPE, water/ice
- Inner liner**  
Food-grade HDPE (IS 10146) [7], smooth
- PUF insulation**  
≈40 mm,  $k \approx 0.022\text{--}0.035 \text{ W/m}\cdot\text{K}$  [6]
- Outer shell**  
UV-stabilised rotomoulded PE
- Sensor (optional)**  
Probe → LED; ESP32 logger optional
- Handles**  
Moulded-in, low centre of gravity

**Made by:** rotomould shell + liner → PUF pour → assemble

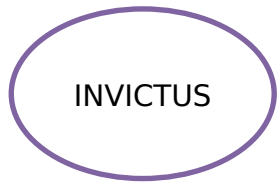
## THERMAL SIMULATION 40 L · 35 °C day



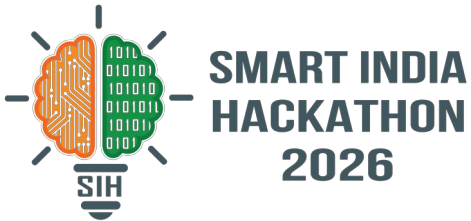
## CARTRIDGES NEEDED (2 L each)

|                         |   |
|-------------------------|---|
| Pre-chilled milk · 12 h | 2 |
| Fresh 30 L · 6 h        | 6 |
| Fresh 40 L · 12 h       | 8 |

**Fresh-milk pull-down is ≈¾ of the load, so cartridge count sets capacity.**  
Sized by simulation: 35 °C ambient, 40 mm PUF; milk cp, ρ [8].



# FEASIBILITY AND VIABILITY



| Analysis of the feasibility of the idea                               | Potential challenges and risks                                | Strategies for overcoming these challenges                |
|-----------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------|
| <b>Cooling duration</b><br>Simulated: 14 h below 8 °C                 | Fresh milk needs $\approx 3\times$ the cooling of a 12 h hold | Cartridge count sized per load by simulation (2–8)        |
| <b>Heat leakage</b><br>$\approx 22$ W at 35 °C (wall heat-flow model) | Lid joint is the main weak spot ( $\approx 11$ % of leak)     | Insulated lid, gasket, no metal through-paths             |
| <b>Weight</b><br>Manageable                                           | $\leq 16$ kg of ice for a full 40 L, 12 h trip                | Carry only what the load needs; swap at collection centre |
| <b>Hygiene</b><br>Food-grade HDPE, easy to clean                      | Milk residue, little cleaning water                           | Smooth liner, wide mouth, crown lifts out                 |
| <b>Leakage · rough roads</b><br>Robust                                | Bumps, tipping, cartridge or lid leaks                        | Rotomoulded PE, gasket + latches, welded cartridges       |
| <b>Cost · manufacturing</b><br>Viable                                 | Tooling cost at volume                                        | Prototype from off-the-shelf drums; rotomould + PUF later |

**TECHNICALLY FEASIBLE**  
Simulation + prior art [1]

**ECONOMICALLY FEASIBLE**  
Prototype  $\approx$  ₹6–13 k (est.)

**OPERATIONALLY FEASIBLE**  
Cartridges swapped at centre

**MANUFACTURABLE**  
Rotomould + PUF process

**SCALABLE**  
Cartridge count sets capacity

## TARGET USERS

- 1 Small & marginal dairy farmers
- 2 Village milk collection centres
- 3 Dairy cooperatives
- 4 Rural milk transporters
- 5 Remote / hilly / NE dairy networks



Today: uninsulated cans. Karnal, Haryana. Photo: Arne Hückelheim, [CC BY-SA 3.0](https://commons.wikimedia.org/wiki/File:Arne_Hueckelheim_-_Karnal_Haryana_India_2014.jpg), Wikimedia Commons

Codex: cool to  $\leq 6$  °C ( $\leq 4$  °C if not collected daily) when not collected within 2 h [4]

## BENEFITS

### SOCIAL

- Safe milk without electricity
- Suits hilly & NE villages

### ECONOMIC

- Reusable ice/PCM, no power bill
- Fewer rejected milk lots

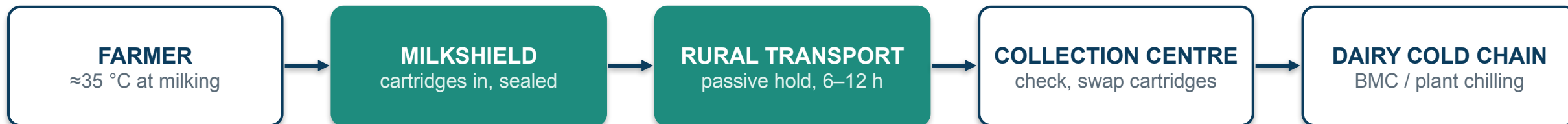
### OPERATIONAL

- Easy to clean, moulded handles
- Cartridges sized per trip

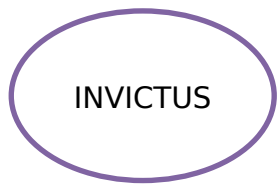
### ENVIRONMENTAL

- No refrigerant, no diesel
- Reusable cartridges, recyclable HDPE

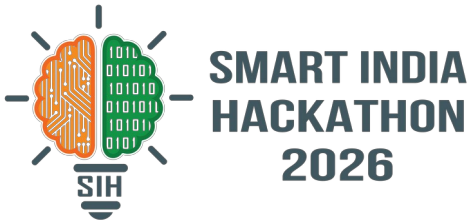
## MILK JOURNEY



Scales later with the same design: 30–40 L → 50–100 L modular → cooperative fleets



# RESEARCH AND REFERENCES



|                 | Conventional can [5] | Ice-in-can prototype [1]                      | PCM pail, ICAR-NDRI [2,3,9]         | MilkShield (proposed)                               |
|-----------------|----------------------|-----------------------------------------------|-------------------------------------|-----------------------------------------------------|
| Cooling         | None                 | Ice compartment                               | Nano-PCM jacket                     | Ice/PCM cartridges in milk                          |
| Power           | None                 | None in use                                   | None in use                         | None in use                                         |
| Reported result | No cooling           | 30 L → 17 °C in <90 min; 20 L <13 °C for 12 h | 37 → <10 °C in 30–40 min (reported) | Simulated: below 8 °C in 75 min; held 14 h at 35 °C |
| Modular         | No                   | No                                            | No                                  | Yes: 2–8 cartridges                                 |
| Cost            | ≈ ₹2,350 (40 L Al)   | Not reported                                  | ≈ ₹1,500–2,000                      | Prototype ≈ ₹6–13 k (est.)                          |

## REFERENCES

[1] Torres-Toledo et al., Int. J. Refrigeration 90 (2018) 22–31. [doi:10.1016/j.ijrefrig.2018.04.001](https://doi.org/10.1016/j.ijrefrig.2018.04.001)

[2] Prakash et al., Innov. Food Sci. Emerg. Technol. 81 (2022) 103112. [doi:10.1016/j.ifset.2022.103112](https://doi.org/10.1016/j.ifset.2022.103112)

[3] Prakash & Ravindra, J. Food Eng. 337 (2023) 111223. [doi:10.1016/j.jfoodeng.2022.111223](https://doi.org/10.1016/j.jfoodeng.2022.111223)

[4] FAO/WHO Codex, Code of Hygienic Practice for Milk and Milk Products, CAC/RCP 57-2004

[5] IGNOU / eGyanKosh, Dairy Technology Unit 1: Milk Collection & Transportation

[6] Verma & Singh, Energy Procedia 161 (2019) 232–241. [doi:10.1016/j.egypro.2019.02.086](https://doi.org/10.1016/j.egypro.2019.02.086)

[7] FSSAI Packaging Regulations 2018 (IS 10146); BIS IS 1825:1983

[8] Minim et al., J. Chem. Eng. Data 47 (2002) 1488. [doi:10.1021/je025546a](https://doi.org/10.1021/je025546a)

[9] DST, Govt. of India: affordable milk chilling at point of harvest

[10] Project repository: [github.com/Sanmugavel1/MilkShield-SIH2026](https://github.com/Sanmugavel1/MilkShield-SIH2026)

**Simulation study:** thermal model · wall heat-flow model · 2D CFD (milk cools evenly, top to bottom within 2.5 °C). Next: prototype field trial.  
**Full simulation code, results & videos:** [github.com/Sanmugavel1/MilkShield-SIH2026](https://github.com/Sanmugavel1/MilkShield-SIH2026)